# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2019

# CHEMISTRY

Past Paper 2019 • Paper Code: B

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION - C (MULTIPLE CHOICE QUESTIONS)

**Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** B

Note: Q. 16: Choose the correct answer from the given options and write the same in your answer copy. Note: Cutting and Over writing is not allowed.

Q1. Benzoic acid is oxidized product of which compound by $KMnO_4$ solution.

(A) Benzene
(B) Benzoyl chloride
(C) Toluene
(D) Cumene

Q2. Which type of hydride is formed when hydrogen reacts with cadmium

(A) Metallic
(B) Ionic
(C) Polymeric
(D) Border line

Q3. When Potassium reacts with oxygen then oxide formed is:

(A) Normal oxide
(B) Peroxide
(C) Super oxide
(D) Sub oxide

Q4. Which of the following is Liquid Metal?

(A) Bromine
(B) Water
(C) Hg
(D) Na

Q5. Enzyme pepsin belongs to which class:

(A) Oxido-Reductase
(B) Lyase
(C) Hydrolase
(D) Ligase

Q6. When one molecule of ethyl alcohol is added in formaldehyde then forming product is:

(A) Oxime
(B) Acetal
(C) Hemiacetal
(D) Cyanohydrin

Q7. The maximum oxidation Number of manganese in the compounds is

(A) + 2
(B) + 4
(C) + 7
(D) + 8

Q8. Which metal does not react with nitric acid?

(A) Cu
(B) Al
(C) Zn
(D) Mg

Q9. Which of the following substance is used as binder in paints?

(A) Turpentine oil
(B) Glue
(C) Linseed oil
(D) Pigment

Q10. The complex lipid in the following is:

(A) Fat
(B) Wax
(C) Cholesterol
(D) Lecithin

Q11. Methyl chloride reacts with aqueous NaOH to form methanol then reaction will be:

(A) $SN_1$
(B) $SN_2$
(C) $E_2$
(D) $E_1$

Q12. Which alkyl halide does not give dehydrohalogenation with alcoholic KOH

(A) $CH_3Cl$
(B) $C_2H_5Cl$
(C) $C_3H_7Br$
(D) $C_4H_9$-I

Q13. Condensation of ethyl alcohol gives the formation of:

(A) Ester
(B) Ether
(C) Ketone
(D) Osazone

Q14. Which of following compound is heated with phenol to form picric acid

(A) Conc: $HNO_3$
(B) Dilute $HNO_3$
(C) $HNO_2$
(D) $H_2SO_4$

Q15. $H_2C_2O_4 + H_2SO_4 \rightarrow CO_2 + CO + H_2O$ Above reaction of sulphuric acid is:

(A) Acidic
(B) Redox
(C) Sulphonation
(D) None of these

Q16. Chlorine reacts with water to form HCl and HOCl, then type of reaction is

(A) Addition
(B) Redox
(C) Substitution
(D) Self redox

Q17. Which one is insoluble in water

(A) $Na_2CO_3$ (B) $CaCl_2$

(C) $Na_2SO_4$ (D) $CaCO_3$

## SECTION – A (SHORT ANSWER QUESTIONS)

**Time Allowed:** 2 Hours and 40 minutes | **Maximum Marks:** 40

Note: NOTE: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. Explain with diagrams and names that which complex is chelate and which one is not chelate. $[Cr(NH_3)_6]\ Cl_3$ and $[Co(en)_3]\ (NO_3)_3$

Q2. Define with chemical reactions that how Bakelite and Nylon are formed?

Q3. Prove that non volatile hydrides are reducing agents and write complete balanced reactions of Potassium hydride with (a) water (b) oil of vitriol (c) Ethanol

Q4. How proteins are classified on the basis of their functions.

Q5. Describe physiochemical principles which are involved in the manufacturing of Sodium carbonate.

Q6. Write the structures of following: (i) Methyl methacrylate (ii) Adipic acid (iii) α – chlorocaporaldehyde

Q7. Explain why Mendeleev's periodic law was rejected.

Q8. Explain the reaction mechanism of nitration and acylation of benzene.

Q9. Describe with the help of any two balanced chemical equations showing that king of compound is dehydrating agent and oxidant.

Q10. Define isomerism and three chain isomers of heptanes with their IUPAC names.

Q11. Show balanced chemical reactions of Sodium hydroxide with the following: (i) CO (ii) $CH_3COONa$ (iii) $Cl_2$ (iv) $CH_3COOC_2H_5$ (v) $NH_4Cl$

Q12. How following reactions are carried out and under what condition? (a) Dehydration of Propyl alcohol (b) Decarboxylation of Sodium benzoate (c) Dehydrohalogenation of vicinal dihalide

## SECTION – B (LONG ANSWER QUESTIONS)

**Maximum Marks:** 28

> Note: NOTE: ATTEMPT ANY TWO (02) QUESTIONS.

## Q13.

(a) [09] Define in detail the Hall's Method, Bayer's Method and Serpek's Method

(b) [05] Describe contact process with steps

## Q14.

(a) [08] Draw the structures of following compounds: (i) Cyanohydrin (ii) Benzene Sulphonate (iii) Hemiacetal (iv) Nicotinic Acid

(b) [06] Name the following compounds: (i) $[Cr(NH_3)_4Cl_2]Cl$ (ii) $K_3[Co(NO_2)_6]$ (iii) S bonded to two $CH_2$–$CH_2$–Cl groups [structure diagram in source scan]

## Q15.

(a) [05] Define simple lipids and their classes with examples.

(b) [05] How Propyl magnesium chloride reacts with: (i) $CO_2$ (ii) Acetone (iii) Formaldehyde

(c) [04] Compare the following properties of “s” block elements. (i) Electrode potential (ii) Hydration enthalpy